

Section 4: Normality Assumption and t -Test

STAT 139 Teaching Team

1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See Canvas for the most accurate information!

1.3 Attendance

- Please notify the Teaching Fellow (TF) if your attendance hasn't been taken!

2 Adding the Normality Assumption

Idea. We began with only four assumptions, but the fifth assumption of *Normality* allows us to extend our inference, especially when we have a limited sample size and cannot appeal to the Central Limit Theorem.

2.1 Point Estimation for Parameters

Definition 1 (ELIH). We begin with the four assumptions of *existence*, *linearity*, *independence*, and *homoskedasticity*:

$$\text{Var}[\varepsilon_i] < \infty,$$

$$\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_i,$$

$$\varepsilon_i \perp\!\!\!\perp \varepsilon_j,$$

and

$$\text{Var}[\varepsilon_i] = \sigma^2$$

for all $i, j \in \{1, \dots, n\}$ and $i \neq j$.

Definition 2 (OLS estimators). $(\hat{\beta}_{0,\text{OLS}}, \hat{\beta}_{1,\text{OLS}}) = \arg \min_{(\beta_0, \beta_1) \in \mathbb{R}^2} (\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2)$, which can be derived under ELIH:

$$\hat{\beta}_{1,\text{OLS}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

and

$$\hat{\beta}_{0,\text{OLS}} = \bar{Y} - \hat{\beta}_{1,\text{OLS}} \bar{x}.$$

Definition 3 (Gauss-Markov Theorem). Under ELIH, the OLS estimators are *optimal* within the class of linear unbiased estimators.

- I.e., they are BLUE: best linear unbiased estimators.

Definition 4 (ELIHN). When we add *normality*, the five main assumptions can be summarized as

$$\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\sigma^2 < \infty$ for all $i \in \{1, \dots, n\}$. Equivalently,

$$Y_i \stackrel{\text{ind.}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2).$$

- Normality is *not* needed for OLS point estimation but gives exact finite-sample distributions for *hypothesis tests* and *confidence intervals*.¹

Definition 5 (OLS-MLE connection). Under ELIHN, the OLS estimators for β_0 and β_1 are the maximum likelihood estimators (MLEs):

$$\hat{\beta}_{1,\text{MLE}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

and

$$\hat{\beta}_{0,\text{MLE}} = \bar{Y} - \hat{\beta}_{1,\text{MLE}} \bar{x}.^2$$

- However,

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2,$$

which is *not* equal to

$$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.^3$$

 **Concept Checker 1.** Which estimator is unbiased: $\hat{\sigma}_{\text{MLE}}^2$ or $\hat{\sigma}_{\text{OLS}}^2$?

Solution

$\hat{\sigma}_{\text{OLS}}^2$ is an unbiased estimator while $\hat{\sigma}_{\text{MLE}}^2$ is not. Recall that $\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-2}^2$. Thus, $\mathbb{E}[\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2] = n-2$ by expectation of chi-squared, and $\mathbb{E}[\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2] = \frac{n-2}{\sigma^2} \mathbb{E}[\hat{\sigma}_{\text{OLS}}^2]$ by linearity, so $\mathbb{E}[\hat{\sigma}_{\text{OLS}}^2] = \sigma^2$ by algebra.

¹Unless otherwise noted, we will assume ELIHN.

²Unless otherwise noted, $\hat{\beta}_1$ refers to $\hat{\beta}_{1,\text{OLS}} = \hat{\beta}_{1,\text{MLE}}$ and $\hat{\beta}_0$ refers to $\hat{\beta}_{0,\text{OLS}} = \hat{\beta}_{0,\text{MLE}}$.

³See Appendix A.1.

2.2 t -Based Inference for Parameters

Definition 6 (Distributions). Under ELIHN, we have the following distributions for our estimators:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right),$$

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}\right)\right),$$

and

$$\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-2}^2.$$

 **Concept Checker 2.** Assume that σ^2 is unknown. Can we use the distribution of $\hat{\beta}_1$ to find a 95% confidence interval for β_1 ?

Solution

We know that $\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right)$, so if σ^2 were known, then we could construct a CI centered at $\hat{\beta}_1$. Using $z_{0.975} = 1.96$, we'd have $\hat{\beta}_1 \pm 1.96 \times \sqrt{\frac{\sigma^2}{s_{xx}}}$. However, we don't know σ^2 , which is why we must plug in an estimator like $\hat{\sigma}_{\text{OLS}}^2$. This changes the distribution to t instead of Normal!

Definition 7 (Hypothesis test for slope). To test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$, we use

$$T = \frac{\hat{\beta}_1 - 0}{\hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{s_{xx}}}} \stackrel{H_0}{\sim} t_{n-2}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

 **Concept Checker 3.** Suppose that instead we wanted to test $H_0 : \beta_1 = 139$ vs. $H_A : \beta_1 \neq 139$. What test statistic would we use? What would its distribution be under H_0 ?

Solution

To test $H_0 : \beta_1 = 139$ vs. $H_A : \beta_1 \neq 139$, we would use $T = \frac{\hat{\beta}_1 - 139}{\hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{s_{xx}}}} \stackrel{H_0}{\sim} t_{n-2}$. Recall that $\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right)$, so $\frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{s_{xx}}} \sim \mathcal{N}(0, 1)$ by standardizing. Additionally, $\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-2}^2$, with $\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \perp\!\!\!\perp \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{s_{xx}}}$. Thus, for all values of β_1 , $\frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{s_{xx}}} \div \sqrt{\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 / (n-2)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{s_{xx}}} \div \frac{\hat{\sigma}_{\text{OLS}}}{\sigma} = \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_{\text{OLS}} \sqrt{1/s_{xx}}} \sim t_{n-2}$ by the story of the t .

Definition 8 (Confidence interval for slope). Our $100(1 - \alpha)\%$ confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{1/s_{xx}}.$$

Definition 9 (Hypothesis test for intercept). To test $H_0 : \beta_0 = 0$ vs. $H_A : \beta_0 \neq 0$, we use

$$T = \frac{\hat{\beta}_0 - 0}{\hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}}} \stackrel{H_0}{\sim} t_{n-2}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

Definition 10 (Confidence interval for intercept). Our $100(1 - \alpha)\%$ confidence interval for β_0 is

$$\hat{\beta}_0 \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}}.$$

 **Concept Checker 4.** We use t_{n-2} (and multiply by $\frac{1}{n-2}$ in the formula for $\hat{\sigma}_{\text{OLS}}^2$) because we lose 2 degrees of freedom. Why? Suppose that we observe $y_1 = 1$ and $y_2 = 1$, with $n = 3$. If we tell you that $\bar{y} = 1$, then what must y_3 be?

Solution

Deterministically, $y_3 = 1$ —it's no longer random, so we lose a degree of freedom. To calculate $\hat{\sigma}_{\text{OLS}}^2$, notice that we must insert values for $\hat{\beta}_0$ and $\hat{\beta}_1$, so we're “double dipping” in the data, losing 2 degrees of freedom.

2.3 Point Estimation for Responses

Definition 11 (Point estimator). Suppose that we're interested in the response Y at some predictor value $X = x_0$. Our *best guess* is

$$\hat{\mu}_{Y|X=x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

2.4 t -Based Inference for Responses

Definition 12 (Population mean confidence interval). We want to find a confidence interval for the *population mean response* at some predictor value—i.e., $\mu_{Y|X=x_0} = \mathbb{E}[Y | X = x_0]$.

- E.g., what is the average height of children whose midparent height is 69.21 inches?
- Our $100(1 - \alpha)\%$ confidence interval for $\mu_{Y|X=x_0}$ is

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}.$$

Definition 13 (Single observation prediction interval). We want to find a prediction interval for a *new response* at some predictor value—i.e., $Y_0 = Y | X = x_0$.

- E.g., what is the predicted height of a child whose midparent height is 69.21 inches?
- Our $100(1 - \alpha)\%$ prediction interval for Y_0 is

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}.$$

- We can decompose the prediction error:

$$\underbrace{Y_0 - \hat{\mu}_{Y|X=x_0}}_{\text{prediction error}} = \underbrace{(Y_0 - \mu_{Y|X=x_0})}_{\text{random sampling error}} + \underbrace{(\mu_{Y|X=x_0} - \hat{\mu}_{Y|X=x_0})}_{\text{estimation error}}.$$

 **Concept Checker 5.** Which has more uncertainty: a population mean confidence interval or a single observation prediction interval?

Solution

The single observation prediction interval has more uncertainty, reflected by the additional +1 in the margin of error. This makes sense since there are so many other factors that go into a new response (e.g., for predicted height of a child, we could look at the nutrition/sleep/lifestyle of the child).

3 *t*-Test as a Linear Model

Idea. We can connect what we learned about the pooled two-sample *t*-test to the simple linear regression model!

Definition 14 (*t*-distribution). Let $Z \sim \mathcal{N}(0, 1)$ and $Y \sim \chi^2_\nu$, with $Z \perp\!\!\!\perp Y$. Then

$$T = \frac{Z}{\sqrt{Y/\nu}} \sim t_\nu.$$

Definition 15 (One-sample *t*-test). Let $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, with μ and σ^2 unknown. Define the *sample variance* as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

We want to conduct the two-sided test $H_0 : \mu = \mu_0$ vs. $H_A : \mu \neq \mu_0$. We use

$$T = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}} \stackrel{H_0}{\sim} t_{n-1}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

Definition 16 (Pooled two-sample *t*-test). Let $Y_{1,1}, \dots, Y_{1,n_1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_1, \sigma^2)$ and $Y_{2,1}, \dots, Y_{2,n_2} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_2, \sigma^2)$, with the two samples independent and μ_1, μ_2 , and σ^2 unknown. Define the *pooled sample variance* as

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

We want to conduct the two-sided test $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$. We use

$$T = \frac{\bar{Y}_1 - \bar{Y}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \stackrel{H_0}{\sim} t_{n_1+n_2-2}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

Definition 17 (Pooled two-sample t -test as a linear model). We can express the setup as

$$Y_{j,i} = \mu_j + \varepsilon_{j,i},$$

where $\varepsilon_{j,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ for all $j \in \{1, 2\}$ and $i \in \{1, \dots, n_j\}$.

- Equivalently,

$$Y_{j,i} = \mu_1 \mathbf{1}\{j = 1\} + \mu_2 \mathbf{1}\{j = 2\} + \varepsilon_{j,i},$$

which can be rewritten in matrix notation as

$$\begin{pmatrix} Y_{1,1} \\ \vdots \\ Y_{1,n_1} \\ Y_{2,1} \\ \vdots \\ Y_{2,n_2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,1} \\ \vdots \\ \varepsilon_{1,n_1} \\ \varepsilon_{2,1} \\ \vdots \\ \varepsilon_{2,n_2} \end{pmatrix}.$$

- As a linear model, let $\beta_0 = \mu_1$ and $\beta_1 = \mu_2 - \mu_1$ such that

$$Y_{j,i} = \beta_0 + \beta_1 \mathbf{1}\{j = 2\} + \varepsilon_{j,i},$$

which can be rewritten in matrix notation as

$$\begin{pmatrix} Y_{1,1} \\ \vdots \\ Y_{1,n_1} \\ Y_{2,1} \\ \vdots \\ Y_{2,n_2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,1} \\ \vdots \\ \varepsilon_{1,n_1} \\ \varepsilon_{2,1} \\ \vdots \\ \varepsilon_{2,n_2} \end{pmatrix}.$$

4 Exercises

 **Problem 1.** For this problem, refer to the code and output below.

- Walk through the code. Specifically, it may help to fill in the comments.
- What are the formulas for `Estimate:(Intercept) = 138.902` and `Estimate:indic = -29.236` from `lm()`?
- Why is `t value:indic = -118.8` from `lm()` different from `t = 115.17` from `t.test()`?

```
```{r}
1:
set.seed(139)

2:
n_1 = 30
```

```

n_2 = 40
mu_1 = 139
mu_2 = 110
sigma = 1

3:
Y_1_vec <- rnorm(n_1, mu_1, sigma)
Y_2_vec <- rnorm(n_2, mu_2, sigma)

4:
Y <- c(Y_1_vec, Y_2_vec)
indic <- c(rep(0, n_1), rep(1, n_2))
df <- as.data.frame(cbind(Y, indic))

5:
t.test(Y_1_vec, Y_2_vec)

6:
summary(lm(Y ~ indic, data = df))
...

```

#### Welch Two Sample t-test

```

data: Y_1_vec and Y_2_vec
t = 115.17, df = 54.401, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 28.72724 29.74491
sample estimates:
mean of x mean of y
 138.902 109.666

Call:
lm(formula = Y ~ indic, data = df)

Coefficients:
 Estimate Std. Error t value Pr(>|t|)
(Intercept) 138.902 0.186 746.8 <2e-16 ***
indic -29.236 0.246 -118.8 <2e-16 ***

```

#### Solution

(a) # 1: Set seed for reproducibility # 2: Define parameters # 3: Generate random data # 4: Define df for lm() # 5: Run t-test # 6: Run linear regression

(b) For `Estimate:(Intercept) = 138.902`, this is  $\bar{y}_1$ . For `Estimate:indic = -29.236`, this is  $\bar{y}_2 - \bar{y}_1$ . In general for linear regression with a categorical predictor in `lm()`, the estimate of the  $y$ -intercept is the sample mean for the baseline group (here,  $\bar{y}_1$ ) and the estimate of the indicator for group  $j$  is the sample mean for group  $j$  minus the sample mean for the baseline group (here,  $\bar{y}_2 - \bar{y}_1$ ).

(c) Importantly, `t.test()`, by default, uses an unpooled  $t$ -test while the standard `lm()` fit assumes a common variance across the two groups and therefore yields the equivalent of the pooled two-sample  $t$ -test (which is appropriate here since the two groups do have shared variance). If we instead ran `t.test(Y_1_vec, Y_2_vec, var.equal = TRUE)`, then we'd see `t = 118.83`. As explained in (b), `lm()` flips the order, so `t = 118.83` from `t.test()` corresponds to `t value:indic = -118.8` from `lm()`.

 **Problem 2.** Let's derive the population mean confidence interval and single observation prediction interval, which are both centered around  $\hat{\mu}_{Y|X=x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ .

(a) Find  $\mathbb{E}[\hat{\mu}_{Y|X=x_0}]$ .

(b) Find  $\text{Cov}[\hat{\beta}_0, \hat{\beta}_1]$ .

(c) Find  $\text{Var}[\hat{\mu}_{Y|X=x_0}]$ .

(d) Find the distribution of  $\hat{\mu}_{Y|X=x_0}$ .

(e) Recall that  $Y_i \stackrel{\text{ind.}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$ . Thus,  $Y_0 \sim \mathcal{N}(\beta_0 + \beta_1 x_0, \sigma^2)$ . This is a new response, so  $Y_0$  is independent of  $Y_1, \dots, Y_n$ —i.e., the data used to calculate  $\hat{\beta}_0$ ,  $\hat{\beta}_1$ , and  $\hat{\sigma}_{\text{OLS}}^2$ . Recall that the prediction error is  $Y_0 - \hat{\mu}_{Y|X=x_0}$ , whose expectation is 0. Find  $\text{Var}[Y_0 - \hat{\mu}_{Y|X=x_0}]$ .

### Solution

(a)

$$\begin{aligned} \mathbb{E}[\hat{\mu}_{Y|X=x_0}] &= \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x_0] && \text{by substituting} \\ &= \mathbb{E}[\hat{\beta}_0] + x_0 \mathbb{E}[\hat{\beta}_1] && \text{by linearity} \\ &= \beta_0 + \beta_1 x_0 && \text{by unbiasedness of OLS estimators} \end{aligned}$$



(b)

$$\begin{aligned}
\text{Cov}[\hat{\beta}_0, \hat{\beta}_1] &= \text{Cov}[\bar{Y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1] && \text{by substituting} \\
&= \text{Cov}[\bar{Y}, \hat{\beta}_1] + \text{Cov}[-\hat{\beta}_1 \bar{x}, \hat{\beta}_1] && \text{by distributing} \\
&= \text{Cov}[\bar{Y}, \hat{\beta}_1] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by bilinearity} \\
&= \text{Cov} \left[ \frac{1}{n} \sum_{i=1}^n Y_i, \frac{\sum_{j=1}^n (x_j - \bar{x})(Y_j - \bar{Y})}{s_{xx}} \right] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by substituting} \\
&= \text{Cov} \left[ \frac{1}{n} \sum_{i=1}^n Y_i, \frac{\sum_{j=1}^n (x_j - \bar{x}) Y_j}{s_{xx}} \right] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{since } \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^n (x_i - \bar{x}) Y_i \\
&= \left( \frac{1}{n s_{xx}} \right) \text{Cov} \left[ \sum_{i=1}^n Y_i, \sum_{j=1}^n (x_j - \bar{x}) Y_j \right] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by bilinearity} \\
&= \left( \frac{1}{n s_{xx}} \right) \sum_{i=1}^n \sum_{j=1}^n \text{Cov} [Y_i, (x_j - \bar{x}) Y_j] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by distributing} \\
&= \left( \frac{1}{n s_{xx}} \right) \sum_{i=1}^n \sum_{j=1}^n (x_j - \bar{x}) \text{Cov} [Y_i, Y_j] - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by bilinearity} \\
&= \left( \frac{1}{n s_{xx}} \right) \left( \sum_{i=j} (x_j - \bar{x}) \text{Cov} [Y_i, Y_j] + \sum_{i \neq j} (x_j - \bar{x}) \text{Cov} [Y_i, Y_j] \right) - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by separating summation} \\
&= \left( \frac{1}{n s_{xx}} \right) \left( \sigma^2 \sum_{i=j} (x_j - \bar{x}) + 0 \sum_{i \neq j} (x_j - \bar{x}) \right) - \bar{x} \text{Var}[\hat{\beta}_1] && \text{by independence} \\
&= \left( \frac{1}{n s_{xx}} \right) (\sigma^2 0) - \bar{x} \text{Var}[\hat{\beta}_1] && \text{since } \sum_{i=1}^n (x_i - \bar{x}) = 0 \\
&= -\bar{x} \text{Var}[\hat{\beta}_1] && \text{by simplifying}
\end{aligned}$$

(c)

$$\begin{aligned}
\text{Var}[\hat{\mu}_{Y|X=x_0}] &= \text{Var}[\hat{\beta}_0 + \hat{\beta}_1 x_0] && \text{by substituting} \\
&= \text{Var}[\hat{\beta}_0] + x_0^2 \text{Var}[\hat{\beta}_1] + 2x_0 \text{Cov}[\hat{\beta}_0, \hat{\beta}_1] && \text{by bilinearity} \\
&= \text{Var}[\hat{\beta}_0] + x_0^2 \text{Var}[\hat{\beta}_1] + 2x_0(-\bar{x} \text{Var}[\hat{\beta}_1]) && \text{by substituting} \\
&= \text{Var}[\hat{\beta}_0] + \text{Var}[\hat{\beta}_1](x_0^2 - 2x_0 \bar{x}) && \text{by simplifying} \\
&= \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right) + \frac{\sigma^2}{s_{xx}} (x_0^2 - 2x_0 \bar{x}) && \text{by substituting} \\
&= \sigma^2 \left( \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} \right) && \text{by simplifying}
\end{aligned}$$

(d)  $\hat{\mu}_{Y|X=x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ , where  $(\hat{\beta}_0, \hat{\beta}_1)$  is Multivariate Normal, so  $\hat{\mu}_{Y|X=x_0}$  is Normal as a linear combination. Thus,  $\hat{\mu}_{Y|X=x_0} \sim \mathcal{N} \left( \beta_0 + \beta_1 x_0, \sigma^2 \left( \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} \right) \right)$ .

(e)

$$\begin{aligned}
\text{Var}[Y_0 - \hat{\mu}_{Y|X=x_0}] &= \text{Var}[Y_0] + \text{Var}[\hat{\mu}_{Y|X=x_0}] - 2\text{Cov}[Y_0, \hat{\mu}_{Y|X=x_0}] && \text{by bilinearity} \\
&= \text{Var}[Y_0] + \text{Var}[\hat{\mu}_{Y|X=x_0}] && \text{by independence} \\
&= \sigma^2 + \sigma^2 \left( \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} \right) && \text{by substituting} \\
&= \sigma^2 \left( 1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} \right) && \text{by substituting}
\end{aligned}$$

## A Proofs

### A.1 OLS-MLE Connection

*Proof.* Assume ELIHN. We want to show that  $\hat{\beta}_{1,\text{MLE}} = \hat{\beta}_{1,\text{OLS}}$ ,  $\hat{\beta}_{0,\text{MLE}} = \hat{\beta}_{0,\text{OLS}}$ , and  $\hat{\sigma}_{\text{MLE}}^2 \neq \hat{\sigma}_{\text{OLS}}^2$ . Recall that  $\hat{\beta}_{1,\text{OLS}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ ,  $\hat{\beta}_{0,\text{OLS}} = \bar{Y} - \hat{\beta}_{1,\text{OLS}}\bar{x}$ , and  $\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$ .

By definition, the maximum likelihood estimators maximize the likelihood function—i.e.,  $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} \mathcal{L}(\vec{\theta}; \vec{Y})$ . Here,  $\vec{\theta} = (\beta_0, \beta_1, \sigma^2)$ , and  $Y_i \stackrel{\text{ind.}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$ .

$$\begin{aligned}
 \mathcal{L}(\beta_0, \beta_1, \sigma^2; \vec{Y}) &= f_{\vec{Y}}(\vec{Y}; \beta_0, \beta_1, \sigma^2) && \text{by def. of likelihood} \\
 &= \prod_{i=1}^n f_{Y_i}(Y_i; \beta_0, \beta_1, \sigma^2) && \text{by independence} \\
 &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(Y_i - \beta_0 - \beta_1 x_i)^2\right) && \text{by Normal PDF} \\
 &= \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2\right) && \text{by multiplying}
 \end{aligned}$$

$$\begin{aligned}
 \ell(\beta_0, \beta_1, \sigma^2; \vec{Y}) &= \log \mathcal{L}(\beta_0, \beta_1, \sigma^2; \vec{Y}) && \text{by def. of log-likelihood} \\
 &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 && \text{by simplifying}
 \end{aligned}$$

For any fixed  $\sigma^2 > 0$ , the first two terms do not depend on  $\beta_0$  or  $\beta_1$ , and  $\frac{1}{2\sigma^2} > 0$ . Thus, *maximizing* the log-likelihood with respect to  $(\beta_0, \beta_1)$  is equivalent to *minimizing*

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2,$$

which is exactly the OLS objective. Therefore,

$$\hat{\beta}_{1,\text{MLE}} = \hat{\beta}_{1,\text{OLS}}$$

and

$$\hat{\beta}_{0,\text{MLE}} = \hat{\beta}_{0,\text{OLS}}.$$

Finally, let's maximize  $\ell(\beta_0, \beta_1, \sigma^2; \vec{Y})$  with respect to  $\sigma^2$ . Let  $c = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$ , which does not depend on  $\sigma^2$ .

$$\begin{aligned}
\frac{\partial}{\partial \sigma^2} \ell(\beta_0, \beta_1, \sigma^2; \vec{Y}) &= \frac{\partial}{\partial \sigma^2} \left( -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} c \right) && \text{by substituting} \\
&= -\frac{n}{2} \times \frac{\partial}{\partial \sigma^2} (\log(\sigma^2)) - \frac{c}{2} \times \frac{\partial}{\partial \sigma^2} \left( \frac{1}{\sigma^2} \right) && \text{by linearity} \\
&= -\frac{n}{2\sigma^2} + \frac{c}{2(\sigma^2)^2} && \text{by simplifying}
\end{aligned}$$

$$\begin{aligned}
0 = -\frac{n}{2\sigma^2} + \frac{c}{2(\sigma^2)^2} &\implies \frac{n}{2\sigma^2} = \frac{c}{2(\sigma^2)^2} && \text{by adding} \\
&\implies n2(\sigma^2)^2 = 2\sigma^2 c && \text{by cross-multiplying} \\
&\implies \sigma^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 && \text{by simplifying}
\end{aligned}$$

Therefore,

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2,$$

which is *not* equal to

$$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

□